Tutorial 4

Difference equations and Laplace transforms

SC3102: Signals, Systems, and Transforms

Question 1

Difference Equations with Initial State

Recall: Linear Constant Coefficient Difference Equation

- LCCDEs are linear relationships between input and output signals
- For example,

$$y[n] + y[n-1] + \frac{1}{3}y[n-2] = x[n+1] - x[n], \qquad n \geq 0.$$

- LCCDEs specify **when** the equation is valid
- Given: $x[n]$ and initial values $y[-1], y[-2]$
- Goal: Compute **future** values of y

Recall: Zero-input and zero-state responses

- The solution y to an LCCDE can be decomposed into:
  (a) **Zero-input Response:** Component of response due to *initial conditions*
  (b) **Zero-state Response:** Component of response due to *input signal*
- The **complete response** is the sum

$$\boxed{y[n] = y_{\text{zi}}[n] + y_{\text{zs}}[n].}$$

Recall: Unilateral Z-transform

- For a real or complex discrete-time signal $x[n]$:

$$X_+(z) = \mathcal{Z}_+\{x[n]\} = \sum_{n=0}^{\infty} x[n]z^{-n}.$$

- Accounts for initial values through the shift formulas.

Recall: Unilateral shift formulas

- One-sample delay:

$$\mathcal{Z}_+\{x[n-1]\} = z^{-1}X_+(z) + x[-1].$$

- Two-sample delay:

$$\mathcal{Z}_+\{x[n-2]\} = z^{-2}X_+(z) + x[-1]z^{-1} + x[-2].$$

Question 1(a): Responses with Initial Conditions

- We are given the LCCDE:

$$y[n] + \tfrac{1}{4}y[n-2] = 3\cos\left(\frac{\pi n}{2}\right) u[n], \qquad n \geq 0,$$

  with $y[-1] = 0$ and $y[-2] = 4$.
- Use the unilateral Z-transform
- Compute zero-input, zero-state and complete responses.
- *Hint:* Partial fractions in z^2 and the cosine pair.

1(a): Transform the LCCDE

- We are given the LCCDE:

$$y[n] + \frac{1}{4}y[n-2] = 3\cos\left(\frac{\pi n}{2}\right) \qquad n \geq 0$$

- By using the (unilateral) delay rule, we get:

$$Y_+(z) + \tfrac{1}{4}\left(z^{-2}Y_+(z) + y[-1]z^{-1} + y[-2]\right) = X_+(z).$$

1(a): Initial Values

- Substitute $y[-1] = 0$, $y[-2] = 4$:

$$Y_+(z) + \tfrac{1}{4}\left(z^{-2}Y_+(z) + 0 \cdot z^{-1} + 4\right) = X_+(z).$$

- We get:

$$Y_+(z) + \tfrac{1}{4}z^{-2}Y_+(z) + 1 = X_+(z).$$

- Rearranging, we get:

$$\left(1 + \tfrac{1}{4}z^{-2}\right) Y_+(z) = X_+(z) - 1.$$

1(a): Zero-input response

- For the **zero-input** response, set $X_+ = 0$:

$$\left(1 + \tfrac{1}{4} z^{-2}\right) Y_{\mathrm{zi},+} = -1.$$

- Match the cosine pair:

$$Y_{\mathrm{zi},+}(z) = -\frac{z^2}{z^2 + 1/4}, \qquad |z| > \tfrac{1}{2}.$$

$$\boxed{y_{\mathrm{zi}}[n] = -\left(\tfrac{1}{2}\right)^n \cos\left(\frac{\pi n}{2}\right), \qquad n \geq 0.}$$

1(a): Zero-state Response

- For the **zero-state**, set *initial values* to zero:

$$\left(1+\tfrac{1}{4}z^{-2}\right) Y_{\mathrm{zs},+}(z) = X_+(z) \implies Y_{\mathrm{zs},+}(z) = \frac{z^2}{z^2+1/4} X_+(z).$$

- What is $X_+(z)$?

$$x[n] = 3\cos\left(\frac{\pi}{2}n\right) u[n] \implies X_+(z) = 3\frac{z^2}{z^2+1}$$

1(a): Partial-fraction coefficients

- Substituting:

$$Y_{\mathrm{zs},+}(z) = \frac{z^2}{z^2 + 1/4} X_+(z) = 3 \frac{z^4}{(z^2+1)(z^2+1/4)}$$

- Now, we can write:

$$\frac{3z^4}{(z^2+1)(z^2+1/4)} = 3z^2 \left(\frac{z^2}{(z^2+1)(z^2+1/4)} \right).$$

1(a): Partial-fraction coefficients

- We can now get with partial fractions:

$$3z^2\left(\frac{z^2}{(z^2+1)(z^2+1/4)}\right) = z^2\left(\frac{4}{z^2+1} - \frac{1}{z^2+1/4}\right)$$

- Hence, we can invert the two terms individually:

$$Y_{\text{zs},+} = 4\frac{z^2}{z^2+1} - \frac{z^2}{z^2+1/4}.$$

1(a): Zero-state response

- We can invert the two unilateral Z-transforms:

$$Y_{\text{zs},+} = 4\frac{z^2}{z^2+1} - \frac{z^2}{z^2+1/4}.$$

- Therefore, the zero-state response is:

$$\boxed{y_{\text{zs}}[n] = \left[4 - \left(\tfrac{1}{2}\right)^n\right] \cos\left(\frac{\pi n}{2}\right).}$$

1(a): Complete response

- We have the zero-input and zero-state responses for $n \geq 0$:

$$y_{\text{zi}}[n] = -\left(\tfrac{1}{2}\right)^n \cos\left(\frac{\pi n}{2}\right),$$
$$y_{\text{zs}}[n] = \left[4 - \left(\tfrac{1}{2}\right)^n\right] \cos\left(\frac{\pi n}{2}\right).$$

- Now, add them to get the complete response for $n \geq 0$:

$$\boxed{y[n] = \left[4 - 2\left(\tfrac{1}{2}\right)^n\right] \cos\left(\frac{\pi n}{2}\right).}$$

Question 1(b): Transient and Steady-state

- Identify the transient and steady-state components:

$$y[n] = \left[4 - 2(\tfrac{1}{2})^n\right] \cos\left(\frac{\pi n}{2}\right), \qquad n \geq 0.$$

- **Transient:** Disappears as $n \to \infty$.
- **Steady state:** Persistent long-term behavior.

1(b): Transient and Steady-state

- Transient:

$$\boxed{y_{\mathrm{tr}}[n] = -2(\tfrac{1}{2})^n \cos\left(\frac{\pi n}{2}\right).}$$

- Steady-state:

$$\boxed{y_{\mathrm{ss}}[n] = 4\cos\left(\frac{\pi}{2}n\right), \qquad n \geq 0.}$$

Question 1(c): Eliminating the Transient

- Choose initial values to eliminate the transient component.
- The desired steady-state output is:

$$y[n] = 4\cos\left(\frac{\pi n}{2}\right), \qquad n \geq 0.$$

- What should $y[-1]$ and $y[-2]$ be?

Use the recurrence!

1(c): Initial values for no transient

- Consider our recurrence and desired solution:

$$y[n] + \frac{1}{4}y[n-2] = 3\cos\left(\frac{\pi}{2}n\right), \qquad y[n] = 4\cos\left(\frac{\pi}{2}n\right)$$

- At $n = 0$, require $y[0] = 4$:

$$4 + \tfrac{1}{4}y[-2] = 3 \quad \Rightarrow \quad y[-2] = -4.$$

- At $n = 1$, require $y[1] = 0$:

$$0 + \tfrac{1}{4}y[-1] = 0 \quad \Rightarrow \quad y[-1] = 0.$$

Question 2

LCCDE and LTI Systems

Recall: System functions and Shift Rule

- LCCDE with zero-input response equal to zero are LTI systems
- **System function:** System function is Z-transform of $h[n]$:

$$y = x * h \quad \longleftrightarrow \quad Y(z) = H(z)X(z).$$

- We can compute H with *bilateral* Z-transform of LCCDE
- Bilateral transform and shift rule:

$$x[n-k] \longleftrightarrow z^{-k}X(z).$$

Question 2(a): System Function

- A real discrete-time LTI system satisfies

$$y[n] - y[n-1] - \tfrac{3}{4}y[n-2] = x[n-k], \qquad n \in \mathbb{Z}.$$

- $k = 2$: Delay input by 2; $k = -1$: Advance input by 1.
- Find $H(z)$ for each k using the bilateral Z-transform.

2(a): Apply the shift rule

- For general k, we are given:

$$y[n] - y[n-1] - \frac{3}{4}y[n-2] = x[n-k].$$

- Using the bilateral shift rule, we get:

$$Y(z) - z^{-1}Y(z) - \frac{3}{4}z^{-2}Y(z) = z^{-k}X(z).$$

2(a): Find $H_k(z)$

- Collect the terms in $Y(z)$:

$$\left(1 - z^{-1} - \frac{3}{4}z^{-2}\right) Y(z) = z^{-k}X(z).$$

- Rearranging, we get:

$$H_k(z) = \frac{Y(z)}{X(z)} = \frac{z^{-k}}{1 - z^{-1} - \frac{3}{4}z^{-2}}.$$

- Multiply by z^2 and factorizing:

$$H_k(z) = \frac{z^{2-k}}{z^2 - z - \frac{3}{4}} = \frac{z^{2-k}}{(z + \frac{1}{2})(z - \frac{3}{2})}.$$

2(a): Substitute $k = -1, 2$

- For $k = -1$, the system function is:

$$\boxed{H_{-1}(z) = \frac{z^3}{(z + \frac{1}{2})(z - \frac{3}{2})}.}$$

- For $k = 2$, the system function is:

$$\boxed{H_2(z) = \frac{1}{(z + \frac{1}{2})(z - \frac{3}{2})}.}$$

Question 2(b): Poles, zeros and ROCs

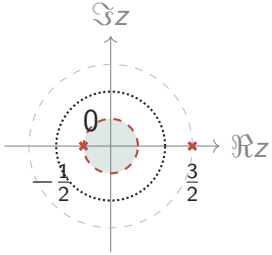
- Plot the poles and zeros; give all possible ROCs:

$$H_2(z) = \frac{1}{(z+\frac{1}{2})(z-\frac{3}{2})}, \qquad H_{-1}(z) = \frac{z^3}{(z+\frac{1}{2})(z-\frac{3}{2})}.$$

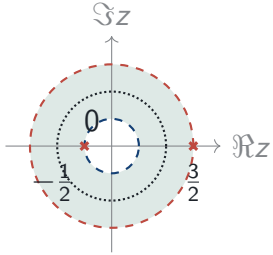
- After cancellation:
  (a) **Poles** are denominator roots
  (b) **Zeros** are numerator roots
- *Pole circles* separate the possible ROCs.

2(b): ROCs for $k = 2$

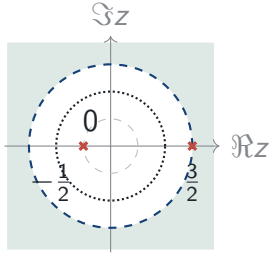
- We have three possible ROCs



$0 < |z| < \frac{1}{2}$



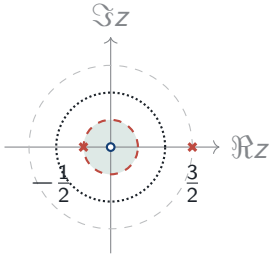
$\frac{1}{2} < |z| < \frac{3}{2}$



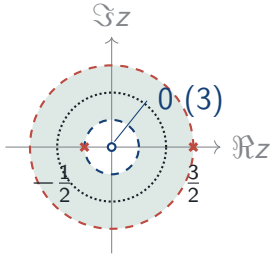
$|z| > \frac{3}{2}$

2(b): ROCs for $k = -1$

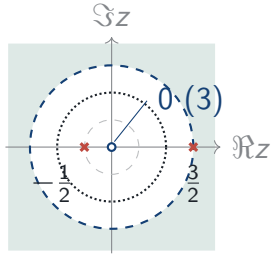
- Again we have three possible ROCs



$0 < |z| < \frac{1}{2}$



$\frac{1}{2} < |z| < \frac{3}{2}$



$|z| > \frac{3}{2}$

Question 2(c–d): Causality and stability

- For each k and each ROC:
- **(c)** Which systems are causal? Which are BIBO stable?
- **(d)** Can any be causal **and** BIBO stable?

2(c): Stability from the ROCs

- BIBO stability requires unit circle in the ROC:

$$\sum_n |h[n]| < \infty \quad \Longleftrightarrow \quad |z| = 1 \text{ lies inside the ROC.}$$

- In the plots, only the annular ROC contains $|z| = 1$:

$$\boxed{\frac{1}{2} < |z| < \frac{3}{2}.}$$

- For both $k = -1$ and $k = 2$: annular is stable ROC

2(c): Causality from the ROCs

- Causality requires:

$$h[n] = 0 \qquad (n < 0).$$

- *If* a causal inverse exists, its ROC must be exterior:

$$\boxed{|z| > \frac{3}{2}.}$$

- We must still *check*!

2(c): Intuition

- For $k = -1$, y uses future input:

$$y[n] = y[n-1] + \frac{3}{4}y[n-2] + x[n+1].$$

- For $k = 2$, it uses a delayed input:

$$y[n] = y[n-1] + \frac{3}{4}y[n-2] + x[n-2].$$

- With exterior ROC, **expect** $k = -1$ to be noncausal and $k = 2$ causal

2(d): Can both conditions hold?

- BIBO stability requires the annular ROC:

$$\frac{1}{2} < |z| < \frac{3}{2}.$$

- Causality would require the exterior ROC:

$$|z| > \frac{3}{2}.$$

- No ROC satisfies both requirements:

No system here is **both** causal and BIBO stable.

Question 2(e): Exterior Impulse Responses

- Take exterior inverses to *verify* causality
- We first remove the k dependent part:

$$H_k(z) = \frac{z^{2-k}}{(z+\frac{1}{2})(z-\frac{3}{2})} = z^{-k}\frac{z^2}{(z+\frac{1}{2})(z-\frac{3}{2})}.$$

- We can invert:

$$\boxed{G(z) = \frac{z^2}{(z+\frac{1}{2})(z-\frac{3}{2})}, \qquad |z| > \frac{3}{2}.}$$

and then use the **shift** property

2(e): Why define G?

- Suppose $g[n] \longleftrightarrow G(z)$
- The bilateral shift rule gives:

$$H_k(z) = z^{-k} G(z) \quad \longleftrightarrow \quad h_k[n] = g[n-k].$$

- Once we know $g[n]$, we can get any h_k:

$$\boxed{h_{-1}[n] = g[n+1], \qquad h_2[n] = g[n-2].}$$

2(e): Inverting $G(z)$

- Compute partial fractions:

$$G(z) = \frac{z^2}{(z+\frac{1}{2})(z-\frac{3}{2})} = \frac{1}{4}\frac{z}{z+\frac{1}{2}} + \frac{3}{4}\frac{z}{z-\frac{3}{2}}.$$

- For the **exterior** ROC:

$$\frac{z}{z-p} \longleftrightarrow p^n u[n].$$

- Applying this, we get:

$$\boxed{g[n] = \left[\frac{1}{4}\left(-\frac{1}{2}\right)^n + \frac{3}{4}\left(\frac{3}{2}\right)^n\right] u[n].}$$

2(e): Delay for $k = 2$

- Substitute $n - 2$ for n in $g[n]$:

$$\boxed{h_2[n] = \left[\frac{1}{4}\left(-\frac{1}{2}\right)^{n-2} + \frac{3}{4}\left(\frac{3}{2}\right)^{n-2}\right] u[n-2].}$$

- The response starts at $n = 2$!
- Hence h_2 is **causal**!

2(e): Advance for $k = -1$

- Substitute $n+1$ for n in the same $g[n]$:

$$h_{-1}[n] = \left[\frac{1}{4}\left(-\frac{1}{2}\right)^{n+1} + \frac{3}{4}\left(\frac{3}{2}\right)^{n+1}\right] u[n+1].$$

- Response starts at $n = -1$!
- Therefore, h_{-1} is **not causal**!

Question 3

Laplace inversion by partial fractions

Recall: Laplace Transform and ROC

- The Laplace transform reweights the input signal and computes CTFT:

$$X(s = \sigma + \mathrm{j}\omega) = \int_{-\infty}^{\infty} \underbrace{x(t)e^{-\sigma t}}_{\text{Re-weighted Signal}} e^{-\mathrm{j}\omega t}\, dt = \int_{-\infty}^{\infty} x(t)e^{-st}\, dt.$$

- Its **region of convergence (ROC)** was s such that:

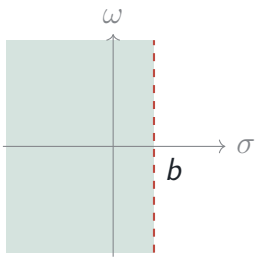
$$\int_{-\infty}^{\infty} |x(t)|\, |e^{-st}|\, dt = \int_{-\infty}^{\infty} |x(t)|e^{-\sigma t}\, dt < \infty.$$

- A Laplace transform is the expression **and** its ROC

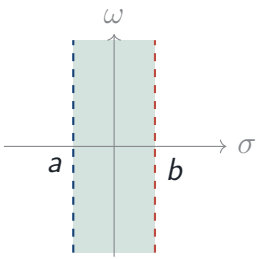
Recall: Possible ROC Shapes

- Laplace transforms have ROCs as vertical bands
- For rational expressions, the **pole lines** are excluded

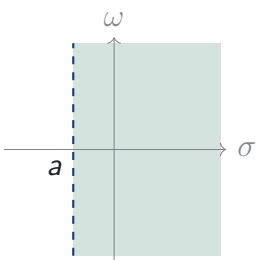
Left-sided ROC



Two-sided ROC



Right-sided ROC



Question 3(a): Repeated Pole

- Invert using partial fractions:

$$A(s) = \frac{s+5}{(s+2)^2}, \qquad \Re s > -2.$$

- Split into known inverse pairs.
- ROC right of $s = -2$: Use **right-sided** pairs.

3(a): Partial fractions

- Since we have **repeated** roots:

$$A(s) = \frac{s+5}{(s+2)^2} = \frac{P}{s+2} + \frac{Q}{(s+2)^2}.$$

- By comparing numerators (after combining denominators):

$$P(s+2) + Q = s + 5 \implies P = 1 \text{ and } Q = 3$$

- Therefore

$$\boxed{A(s) = \frac{1}{s+2} + \frac{3}{(s+2)^2}.}$$

3(a): Taking Inverse

- For $\Re s > -2$, take right-sided inverses for both terms:

$$\frac{1}{s+2} \longleftrightarrow e^{-2t}u(t), \qquad \frac{1}{(s+2)^2} \longleftrightarrow te^{-2t}u(t).$$

- Inverting term by term and adding, we get:

$$\boxed{a(t) = (1+3t)e^{-2t}u(t), \qquad \Re s > -2.}$$

Question 3(b): Denominator with Complex Roots

- Invert using partial fractions:

$$B(s) = \frac{5}{(s+1)(s^2+4)}, \qquad \Re s > 0.$$

- *Hint:* Use the following partial fraction expansion

$$B(s) = \frac{P}{s+1} + \frac{Qs+R}{s^2+4}.$$

- Generally easier to not split quadratics with **complex roots**!

3(b): Partial fractions

- We will match each of the terms with inverses

$$B(s) = \frac{5}{(s+1)(s^2+4)} = \frac{P}{s+1} + \frac{Qs+R}{s^2+4}.$$

- By comparing numerators (after combining denominators):

$$5 = P(s^2+4) + (Qs+R)(s+1).$$

- Matching coefficients, we get:

$$P+Q=0, \qquad Q+R=0, \qquad 4P+R=5 \quad \Rightarrow \quad Q=-1, \quad P=R=1.$$

3(b): Inverse pairs

- Split into standard pairs:

$$B(s) = \frac{1}{s+1} - \frac{s}{s^2+4} + \frac{1}{s^2+4}.$$

- First term has *exponential* inverse
- For the other two, we have the *sinusoidal* inverses

$$\cos(2t)u(t) \leftrightarrow \frac{s}{s^2+4}, \qquad \sin(2t)u(t) \leftrightarrow \frac{2}{s^2+4}, \qquad \Re s > 0.$$

$$\boxed{b(t) = \left[e^{-t} - \cos(2t) + \tfrac{1}{2}\sin(2t)\right] u(t).}$$

Question 3(c): Denominator with Complex Roots - 2

- Invert using partial fractions:

$$C(s) = \frac{5}{(s+3)(s^2+2s+2)}, \qquad \Re s > -1.$$

- *Hint:* Use the following partial fraction expansion

$$C(s) = \frac{P}{s+3} + \frac{Qs+R}{s^2+2s+2}.$$

- Again, easier to not split quadratics with **complex roots**!

3(c): Partial fractions

- We will match each of the terms with inverses

$$C(s) = \frac{5}{(s+3)(s^2+2s+2)} = \frac{P}{s+3} + \frac{Qs+R}{s^2+2s+2}.$$

- By comparing numerators (after combining denominators):

$$5 = P(s^2+2s+2) + (Qs+R)(s+3).$$

- Matching coefficients, we get:

$$P+Q=0, \qquad 2P+3Q+R=0, \qquad 2P+3R=5$$
$$\Rightarrow \quad Q=-1, \quad P=R=1.$$

3(c): Partial Fractions

- Split into partial fractions:

$$C(s) = \frac{1}{s+3} - \frac{s}{s^2+2s+2} + \frac{1}{s^2+2s+2}.$$

- Since the denominator has **complex roots**, complete the square:

$$s^2 + 2s + 2 = (s+1)^2 + 1, \qquad -s + 1 = -(s+1) + 2.$$

- Therefore,

$$C(s) = \frac{1}{s+3} - \frac{s+1}{(s+1)^2+1} + \frac{2}{(s+1)^2+1}.$$

3(c): Inverse pairs

- First term has *exponential* inverse
- For the other two, we have the *damped sinusoidal* inverses

$$e^{-t}\cos(t)u(t) \leftrightarrow \frac{s+1}{(s+1)^2+1},$$
$$e^{-t}\sin(t)u(t) \leftrightarrow \frac{1}{(s+1)^2+1}, \qquad \Re s > -1.$$

$$\boxed{c(t) = \left[e^{-3t} - e^{-t}\cos(t) + 2e^{-t}\sin(t)\right] u(t).}$$

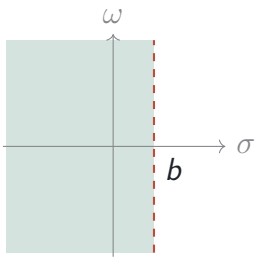
Question 4

Laplace ROCs

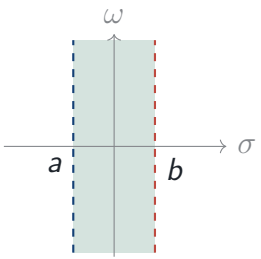
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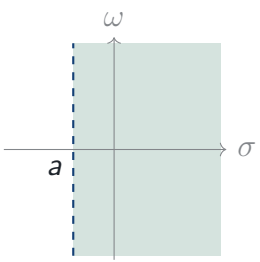
Left-sided ROC



Two-sided ROC



Right-sided ROC



Question 4(a): Poles, zeros and ROCs

- For a continuous-time signal with Laplace transform:

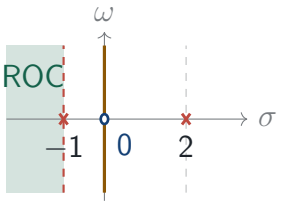
$$F_a(s) = \frac{s - a}{(s + 1)(s - 2)}.$$

- For $a = 0$, sketch the poles, zero and all possible ROCs.

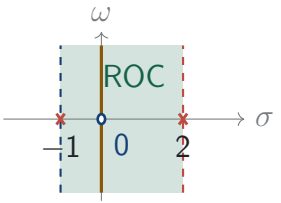
4(a): Three ROCs

- For $a = 0$, F_a reduces to:

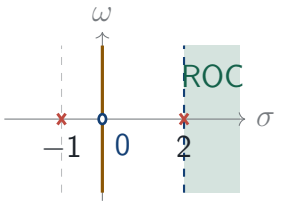
$$F_0(s) = \frac{s}{(s+1)(s-2)}$$



$\sigma < -1$



$-1 < \sigma < 2$



$\sigma > 2$

Question 4(b): Inverses and CTFT

- For $a = 0$, find f_0 for each ROC of:

$$F_0(s) = \frac{s}{(s+1)(s-2)}.$$

- For which of these does the CTFT exist?
- CTFT exists when the **imaginary axis** ($\sigma = 0$) lies in the ROC

4(b): Partial fractions

- We can write the partial fraction expansion of F_0:

$$\frac{s}{(s+1)(s-2)} = \frac{P}{s+1} + \frac{Q}{s-2}.$$

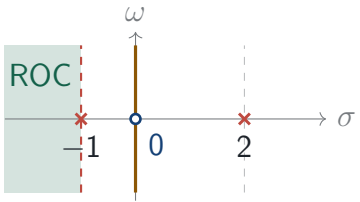
- By comparing numerators:

$$s = P(s-2) + Q(s+1) \quad \Longrightarrow \quad P = \frac{1}{3},\ Q = \frac{2}{3}.$$

- Therefore our partial fraction expansion is:

$$\boxed{F_0(s) = \frac{1}{3}\frac{1}{s+1} + \frac{2}{3}\frac{1}{s-2}.}$$

4(a–b): ROC $\sigma < -1$

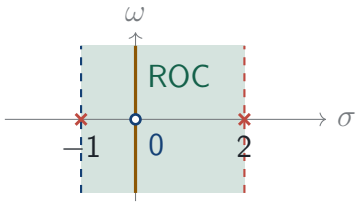


- We will take the left-sided inverse for both terms:

$$\boxed{f_0(t) = -\left(\tfrac{1}{3}e^{-t} + \tfrac{2}{3}e^{2t}\right) u(-t).}$$

- **No CTFT**: $\sigma = 0$ is outside the ROC

4(a–b): ROC $-1 < \sigma < 2$

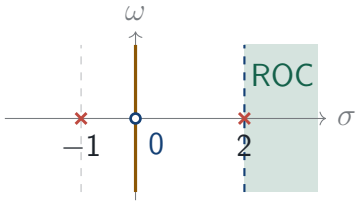


- We will take the right-sided inverse at -1 and left-sided at 2:

$$\boxed{f_0(t) = \tfrac{1}{3}e^{-t}u(t) - \tfrac{2}{3}e^{2t}u(-t).}$$

- **CTFT exists**: $\sigma = 0$ is inside the ROC

4(a–b): ROC $\sigma > 2$



- We will take the right-sided inverse for both terms:

$$\boxed{f_0(t) = \left(\tfrac{1}{3}e^{-t} + \tfrac{2}{3}e^{2t}\right) u(t).}$$

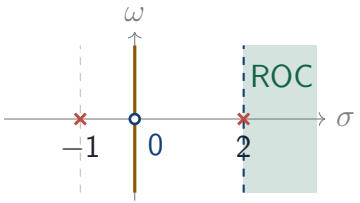
- **No CTFT**: $\sigma = 0$ is outside the ROC

Question 4(c): Right-sided inverse with CTFT

- Given:
$$F_a(s) = \frac{s-a}{(s+1)(s-2)}, \qquad a \in \mathbb{R}.$$
- Which a gives a right-sided inverse with a CTFT?
- Sketch its poles, zeros and ROC.

4(c): Failure of $a = 0$

- The right-sided ROC did not have CTFT was because of **pole** at $s = 2$:



- Can we set a to eliminate the pole?

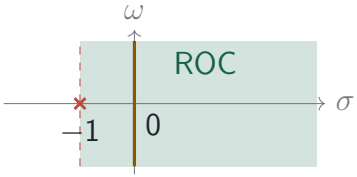
$$F_a(s) = \frac{s - a}{(s + 1)(s - 2)} \implies \boxed{\text{Set } a \text{ to 2!}}$$

4(c): Setting $a = 2$

- At $a = 2$:

$$F_2(s) = \frac{s-2}{(s+1)(s-2)} = \frac{1}{s+1}.$$

- F_2 now has a single pole at $s = -1$:



- **CTFT exists:** New ROC includes $\sigma = 0$!